Literature Review

Machine Learning for Diabetes Classification

Project	Diabetes status classification
Milestone	Literature Review - due 16 October 2026

1. Review Purpose and Scope

This focused review places the team project in the context of published diabetes prediction research. It compares study populations, prediction targets, input variables, model results, and limitations to identify what the current project can reasonably investigate. The supplied dataset contains gender, age, hypertension, heart_disease, smoking_history, bmi, HbA1c_level, blood_glucose_level, and a binary diabetes label.

A key distinction in the literature is whether a model classifies an existing diabetes label or predicts future disease onset. The fields visible in the team dataset do not show a follow-up period, so this project is currently framed as binary classification of the recorded diabetes label. It should not be described as predicting future onset unless the dataset documentation confirms a longitudinal outcome.

2. Evidence from Previous Studies

Xie et al. (2019): Cross-sectional survey classification

Xie and colleagues compared eight machine-learning classifiers using cross-sectional data from 138,146 participants in the 2014 U.S. Behavioral Risk Factor Surveillance System. The models achieved AUC values from 0.7182 to 0.7949. A neural network had the highest accuracy (82.4%) and AUC (0.7949), while a decision tree had the highest sensitivity (51.6%). This difference matters for the team project: a model that scores well on overall accuracy can still miss many diabetes-positive records, so sensitivity/recall should be reported alongside accuracy. The study also notes that cross-sectional, self-reported data cannot establish causality and may be affected by recall bias (Xie et al., 2019).

Farran et al. (2019): Future risk in a Kuwaiti cohort

Farran and colleagues used retrospective health-network data from 1,837 native Kuwaiti Arab adults to estimate diabetes onset over 3-, 5-, and 7-year horizons. Their non-invasive inputs included age, sex, BMI, hypertension, and family-history indicators. k-nearest neighbours (k-NN) achieved AUC values of 0.83, 0.82, and 0.79 across the three horizons, exceeding the reported logistic regression and support vector machine results. Performance declined as the prediction horizon lengthened. The cohort was mainly older and overweight/obese, and the authors lacked several lifestyle and medication variables, limiting how broadly the results can be generalized (Farran et al., 2019).

Lugner et al. (2024): Longitudinal prediction with laboratory features

Using UK Biobank data, Lugner and colleagues modeled incident type 2 diabetes over ten years among 448,277 participants without diabetes at baseline. An XGBoost model ranked HbA1c highest, followed by measures including BMI and blood glucose. The full model reached an ROC-AUC of 0.90 and a reduced ten-feature model 0.88; the full model sensitivity was 0.62. This supports examining HbA1c, glucose, and BMI in the team dataset, while also showing that a high AUC does not guarantee detection of every positive case. The cohort and longitudinal outcome differ from the team's visible snapshot, so the reported AUC cannot be treated as an expected result for this project (Lugner et al., 2024).

3. Critical Synthesis

Across the studies, performance depends on the population, outcome definition, predictor set, and time horizon; no single algorithm is consistently best. Xie et al. show a trade-off between ranking performance and sensitivity, Farran et al. show that k-NN can work well for a specific Kuwaiti cohort but declines over longer horizons, and Lugner et al. show that laboratory markers can dominate a long-term model. These results are not directly comparable because the studies use different cohorts and distinguish current status from future onset.

A systematic review by Fregoso-Aparicio et al. (2021) also describes substantial variation in the algorithms and features reported across type 2 diabetes prediction studies, making it difficult to select a universally optimal model. Taken together, the literature supports a transparent comparison on one clearly described dataset rather than assuming a published model will transfer unchanged.

4. Project-Specific Research Gap

The reviewed studies use different populations and targets: a cross-sectional survey label, future risk in a Kuwaiti cohort, and ten-year incidence in UK Biobank. For the team's supplied dataset, the project-specific question is which selected classifiers perform best on the available combination of demographic, lifestyle, and clinical measurements when predicting its recorded binary diabetes label. The dataset screenshot does not show longitudinal follow-up or external validation, so the project will not claim to forecast future onset or provide a clinically validated tool.

5. Justification and Research Question

This comparison is justified because prior studies report meaningful differences between accuracy, AUC, and sensitivity, and their best-performing algorithms vary by dataset. Testing several established models on the same prepared data will show which approach fits this dataset and whether its positive class is detected reliably.

Research question: How well can Logistic Regression, Decision Tree, and Random Forest classify the diabetes label using the team dataset's demographic, lifestyle, and clinical features?

6. Proposed Conceptual Framework

The project follows this sequence:

- Inputs: gender, age, hypertension, heart disease, smoking history, BMI, HbA1c, and blood glucose.
- Preparation: verify the target and class balance; inspect missing values and duplicates; encode categories and scale numeric inputs when needed.
- Model comparison: use Logistic Regression as a baseline, Decision Tree for an interpretable comparison, and Random Forest for nonlinear patterns.
- Evaluation: use a stratified holdout split and report the confusion matrix, accuracy, precision, recall/sensitivity, F1-score, and ROC-AUC; add cross-validation if the dataset size permits.
- Interpretation: compare errors and feature patterns, describe limitations, and avoid clinical claims.

7. Methodological Checks and Limitations

- Confirm how the diabetes label was created. If HbA1c or blood glucose values were used directly to assign the label, flag possible target leakage or near-direct prediction and interpret results accordingly.
- Check the proportion of positive and negative labels before training. If classes are imbalanced, report recall and F1-score and keep any resampling inside the training data only.
- Use preprocessing fitted on training data only, and use a stratified split so both classes are represented in training and testing.
- The dataset's source, sample size, population, and collection period should be added to the final report after checking its documentation; the screenshot alone does not establish these details.

Scope note: This is an educational data-classification project, not a medical diagnostic system.

References

Farran, B., AlWotayan, R., Alkandari, H., Al-Abdulrazzaq, D., Channanath, A., & Thanaraj, T. A. (2019). Use of non-invasive parameters and machine-learning algorithms for predicting future risk of type 2 diabetes: A retrospective cohort study of health data from Kuwait. Frontiers in Endocrinology, 10, 624. https://doi.org/10.3389/fendo.2019.00624

Fregoso-Aparicio, L., Noguez, J., Montesinos, L., & García-García, J. A. (2021). Machine learning and deep learning predictive models for type 2 diabetes: A systematic review. Diabetology & Metabolic Syndrome, 13, 148. https://doi.org/10.1186/s13098-021-00767-9

Lugner, M., Rawshani, A., Helleryd, E., et al. (2024). Identifying top ten predictors of type 2 diabetes through machine learning analysis of UK Biobank data. Scientific Reports, 14, 2102. https://doi.org/10.1038/s41598-024-52023-5

Xie, Z., Nikolayeva, O., Luo, J., & Li, D. (2019). Building risk prediction models for type 2 diabetes using machine learning techniques. Preventing Chronic Disease, 16, E130. https://doi.org/10.5888/pcd16.190109